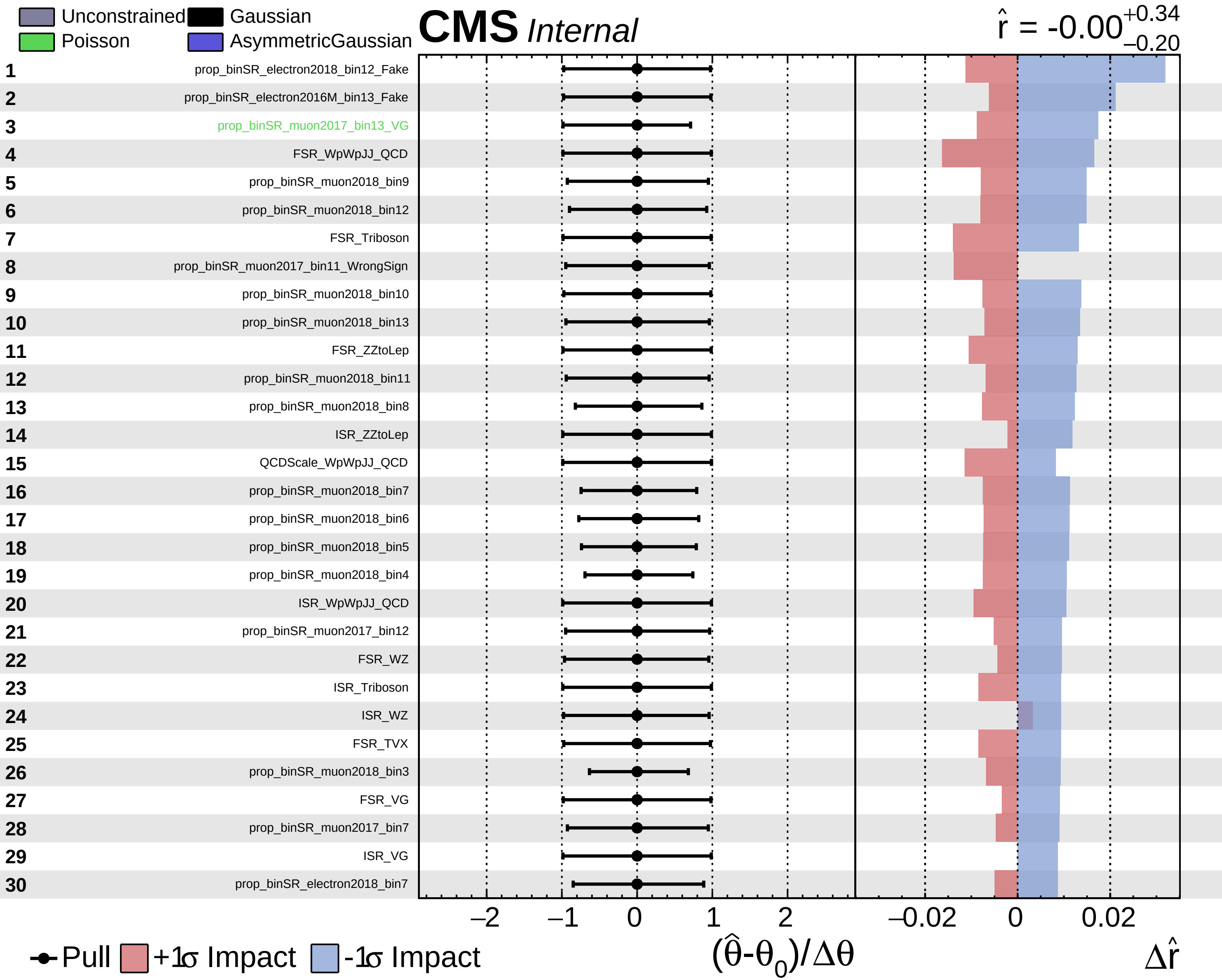
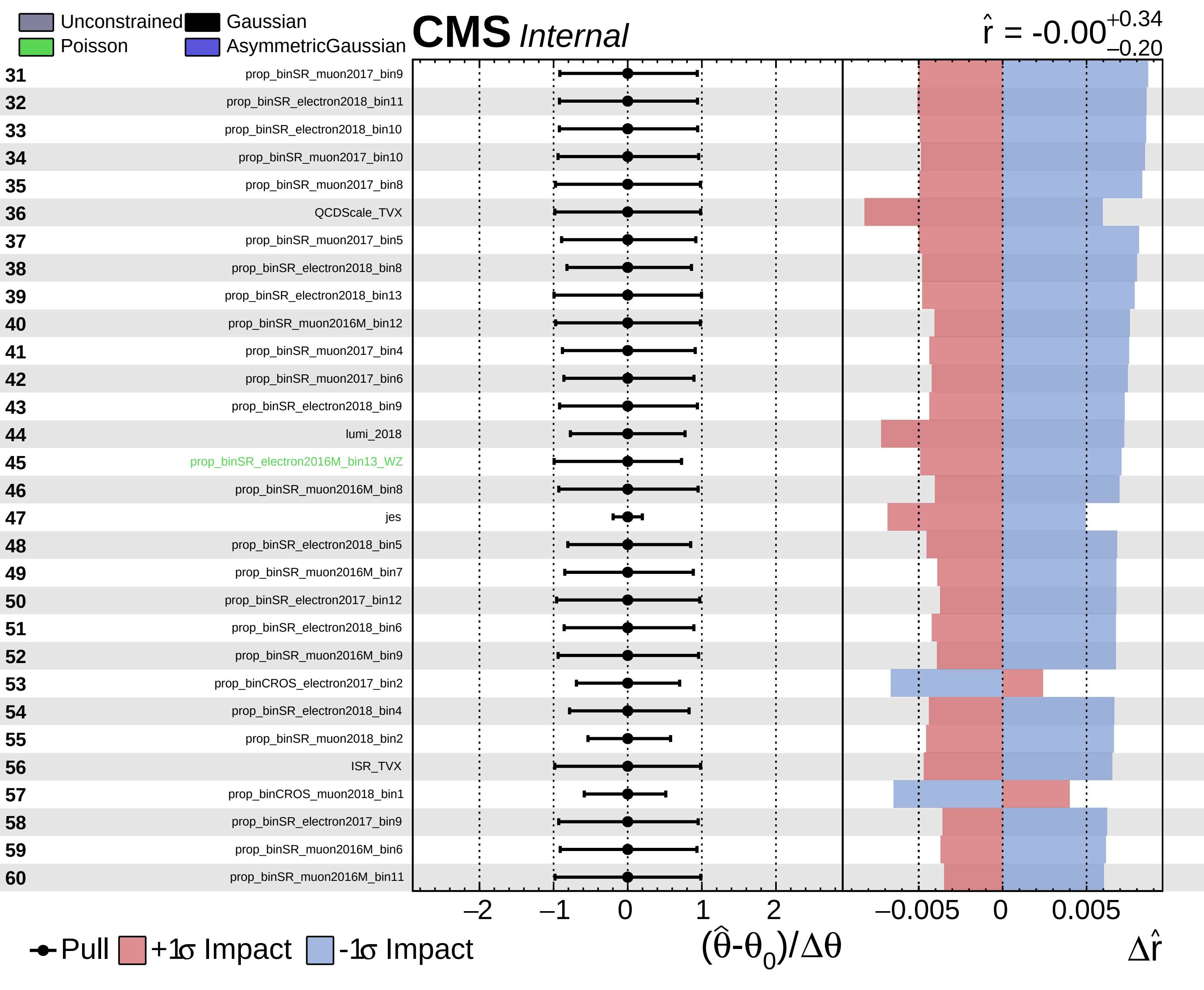
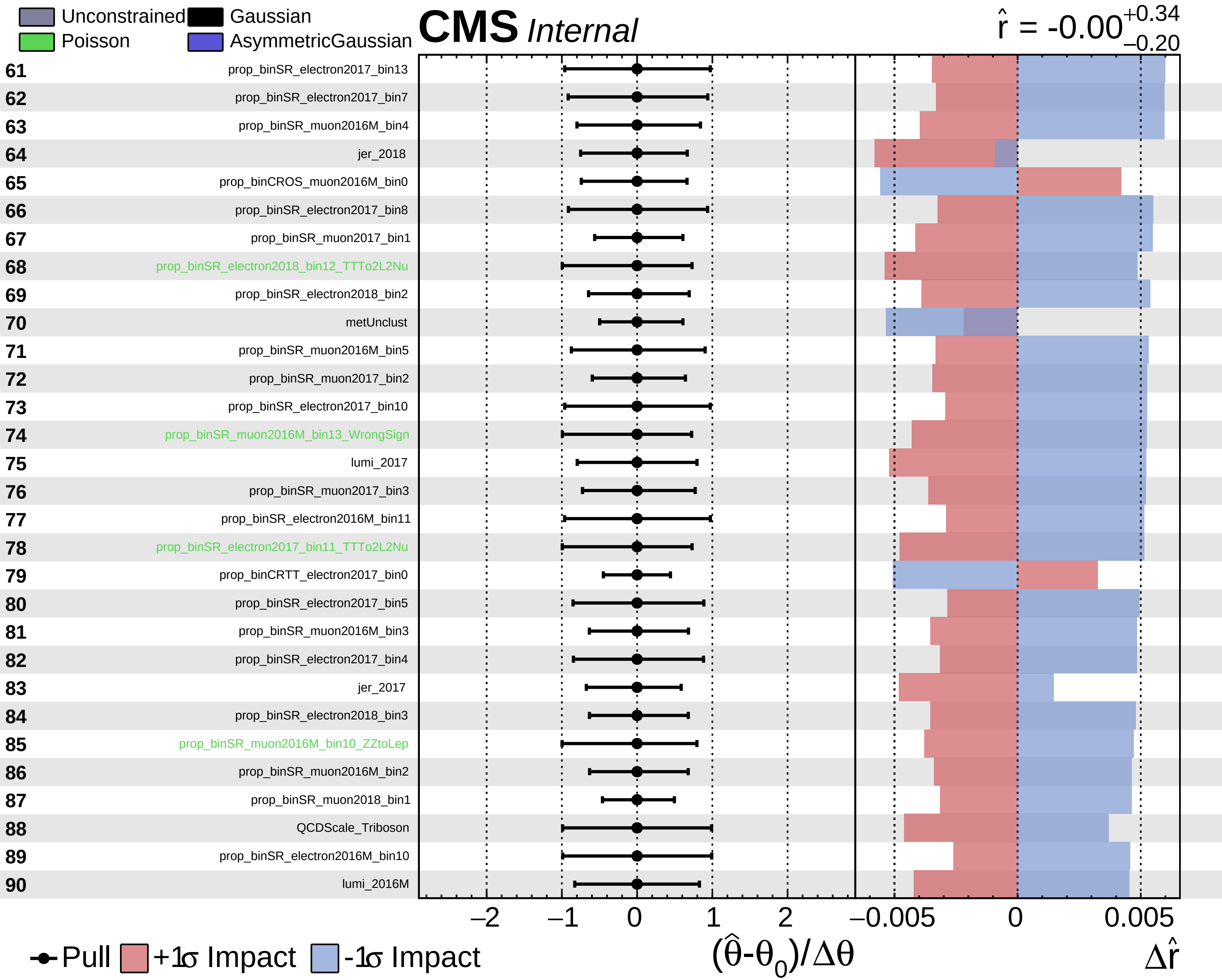
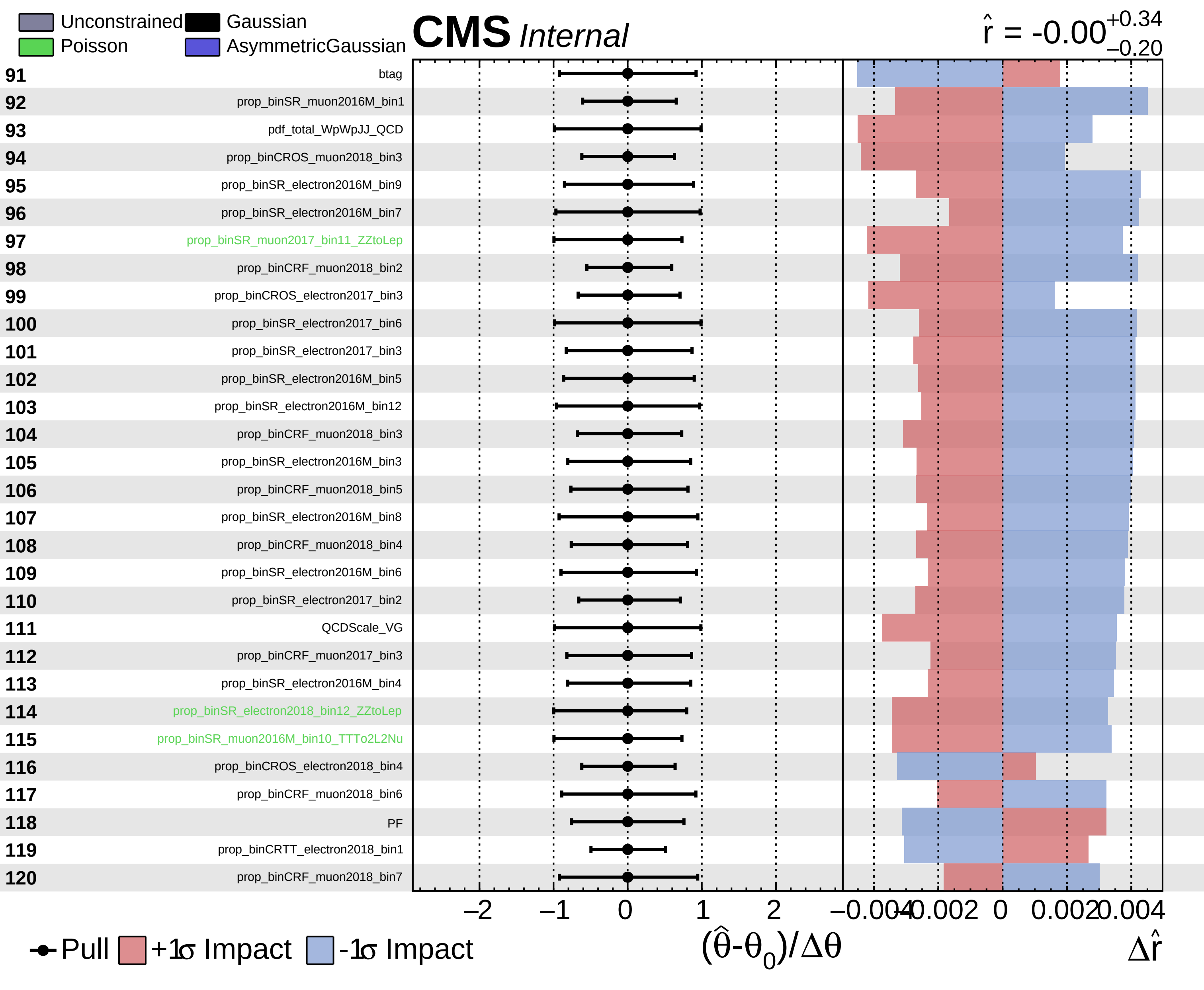

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.34}_{-0.20}$
1 prop_binSR_electron2018_bin12_Fake
2 prop_binSR_electron2016M_bin13_Fake
3 prop_binSR_muon2017_bin13_VG
4 FSR_WpWpJJ_QCD
5 prop_binSR_muon2018_bin9
6 prop_binSR_muon2018_bin12
7 FSR_Triboson
8 prop_binSR_muon2017_bin11_WrongSign
9 prop_binSR_muon2018_bin10
10 prop_binSR_muon2018_bin13
11 FSR_ZZtoLep
12 prop_binSR_muon2018_bin11
13 prop_binSR_muon2018_bin8
14 ISR_ZZtoLep
15 QCDScale_WpWpJJ_QCD
16 prop_binSR_muon2018_bin7
17 prop_binSR_muon2018_bin6
18 prop_binSR_muon2018_bin5
19 prop_binSR_muon2018_bin4
20 ISR_WpWpJJ_QCD
21 prop_binSR_muon2017_bin12
22 FSR_WZ
23 ISR_Triboson
24 ISR_WZ
25 FSR_TVX
26 prop_binSR_muon2018_bin3
27 FSR_VG
28 prop_binSR_muon2017_bin7
29 ISR_VG
30 prop_binSR_electron2018_bin7
−2
−1
0
1
2
−0.02
0
0.02
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

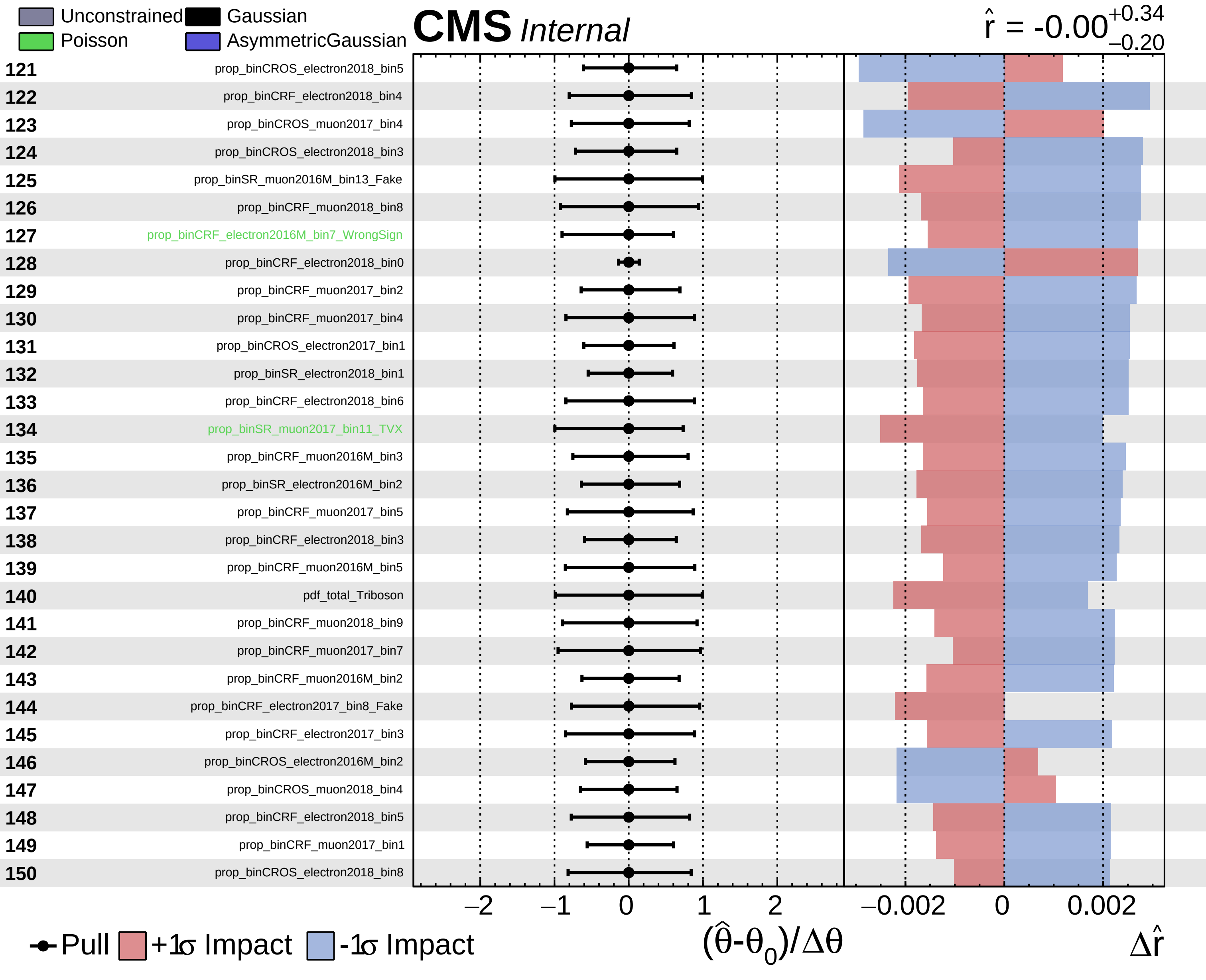

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.34}_{-0.20}$
31 prop_binSR_muon2017_bin9
32 prop_binSR_electron2018_bin11
33 prop_binSR_electron2018_bin10
34 prop_binSR_muon2017_bin10
35 prop_binSR_muon2017_bin8
36 QCDScale_TVX
37 prop_binSR_muon2017_bin5
38 prop_binSR_electron2018_bin8
39 prop_binSR_electron2018_bin13
40 prop_binSR_muon2016M_bin12
41 prop_binSR_muon2017_bin4
42 prop_binSR_muon2017_bin6
43 prop_binSR_electron2018_bin9
44 lumi_2018
45 prop_binSR_electron2016M_bin13_WZ
46 prop_binSR_muon2016M_bin8
47 jes
48 prop_binSR_electron2018_bin5
49 prop_binSR_muon2016M_bin7
50 prop_binSR_electron2017_bin12
51 prop_binSR_electron2018_bin6
52 prop_binSR_muon2016M_bin9
53 prop_binCROS_electron2017_bin2
54 prop_binSR_electron2018_bin4
55 prop_binSR_muon2018_bin2
56 ISR_TVX
57 prop_binCROS_muon2018_bin1
58 prop_binSR_electron2017_bin9
59 prop_binSR_muon2016M_bin6
60 prop_binSR_muon2016M_bin11
−2
−1
0
1
2
−0.005
0
0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



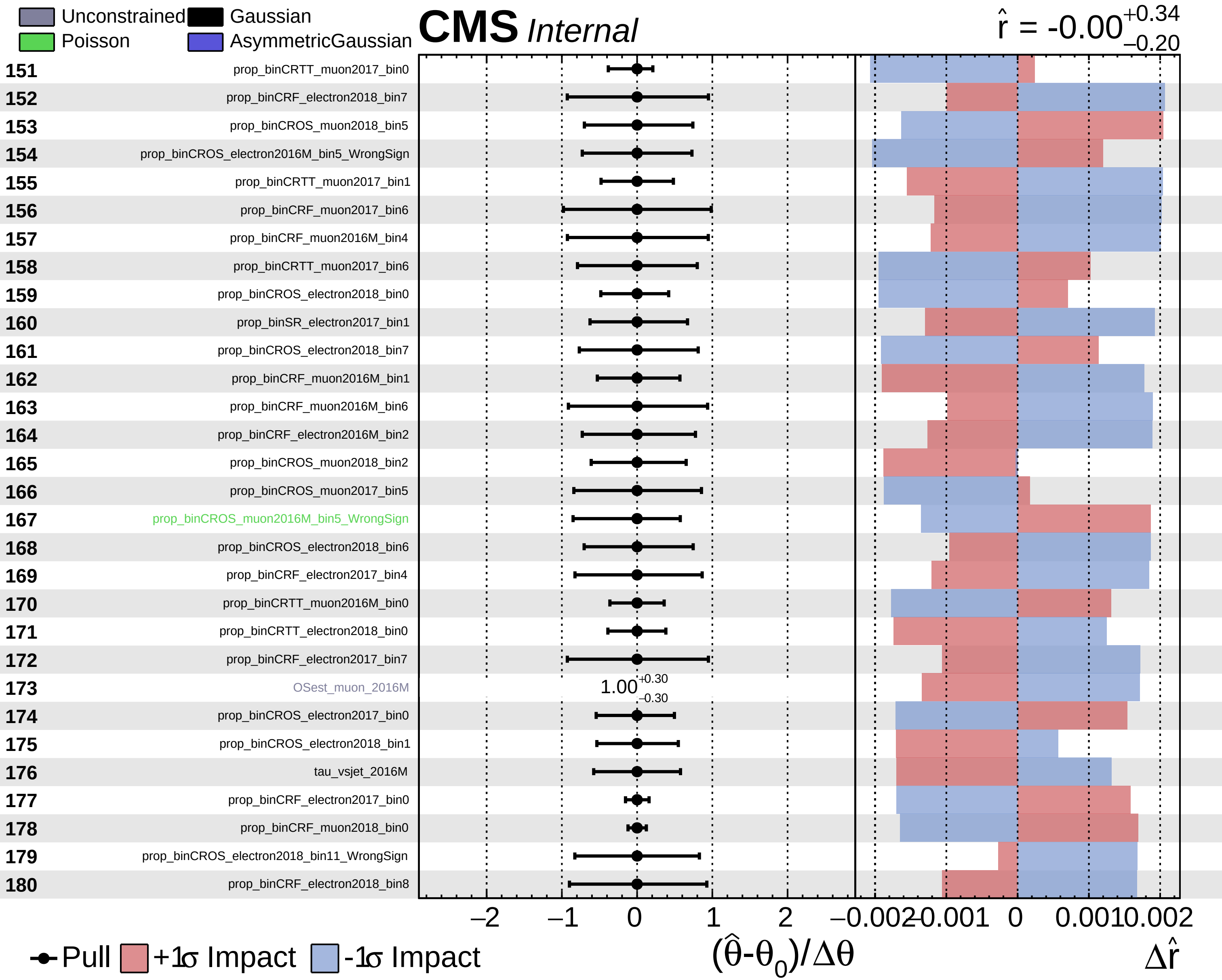
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.34}_{-0.20}$
61 prop_binSR_electron2017_bin13
62 prop_binSR_electron2017_bin7
63 prop_binSR_muon2016M_bin4
64 jer_2018
65 prop_binCROS_muon2016M_bin0
66 prop_binSR_electron2017_bin8
67 prop_binSR_muon2017_bin1
68 prop_binSR_electron2018_bin12_TTTo2L2Nu
69 prop_binSR_electron2018_bin2
70 metUnclust
71 prop_binSR_muon2016M_bin5
72 prop_binSR_muon2017_bin2
73 prop_binSR_electron2017_bin10
74 prop_binSR_muon2016M_bin13_WrongSign
75 lumi_2017
76 prop_binSR_muon2017_bin3
77 prop_binSR_electron2016M_bin11
78 prop_binSR_electron2017_bin11_TTTo2L2Nu
79 prop_binCRTT_electron2017_bin0
80 prop_binSR_electron2017_bin5
81 prop_binSR_muon2016M_bin3
82 prop_binSR_electron2017_bin4
83 jer_2017
84 prop_binSR_electron2018_bin3
85 prop_binSR_muon2016M_bin10_ZZtoLep
86 prop_binSR_muon2016M_bin2
87 prop_binSR_muon2018_bin1
88 QCDScale_Triboson
89 prop_binSR_electron2016M_bin10
90 lumi_2016M
−2 −1 0 1 2
−0.005 0 0.005
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



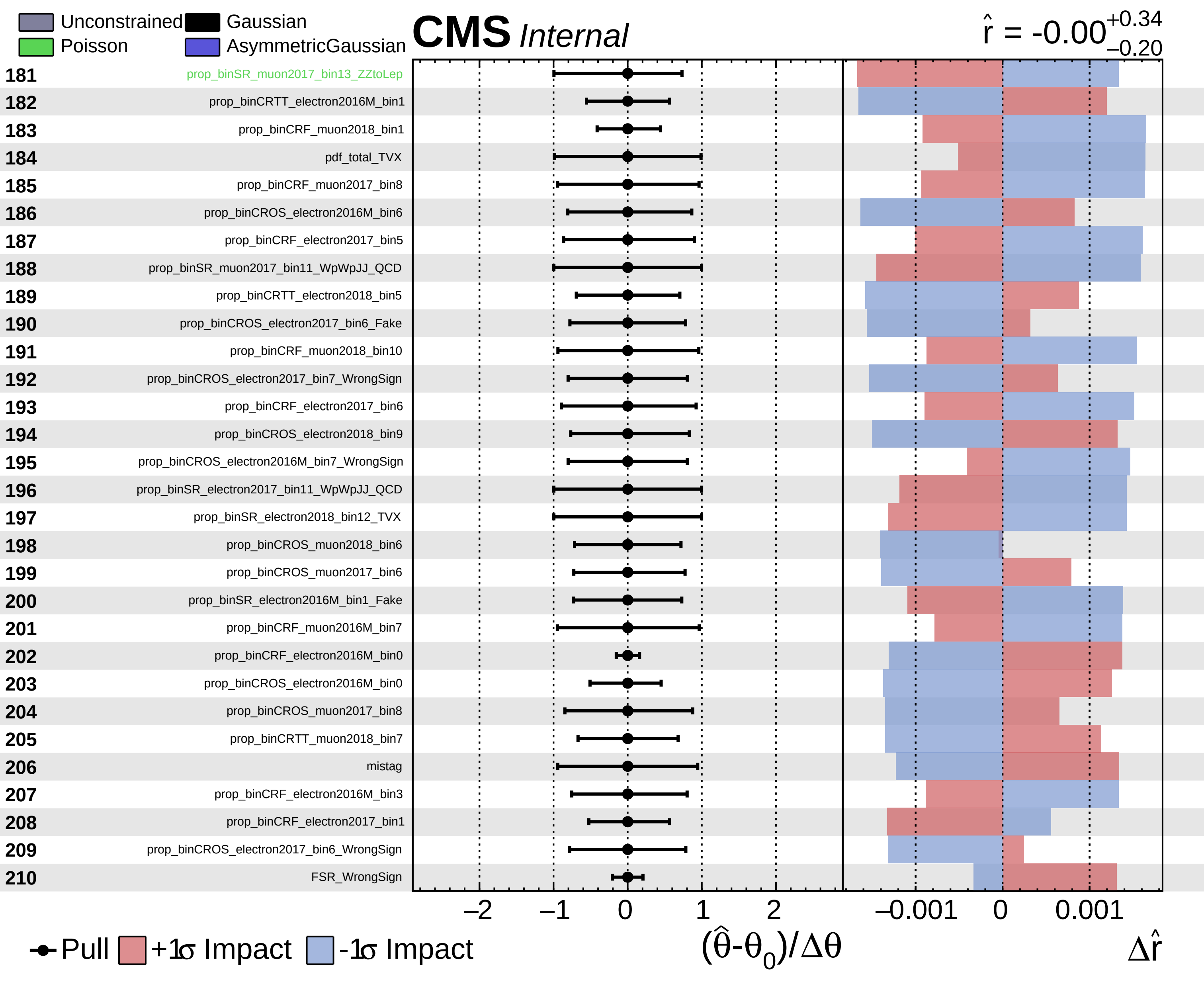
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.34}_{-0.20}$
91 btag
92 prop_binSR_muon2016M_bin1
93 pdf_total_WpWpJJ_QCD
94 prop_binCROS_muon2018_bin3
95 prop_binSR_electron2016M_bin9
96 prop_binSR_electron2016M_bin7
97 prop_binSR_muon2017_bin11_ZZtoLep
98 prop_binCRF_muon2018_bin2
99 prop_binCROS_electron2017_bin3
100 prop_binSR_electron2017_bin6
101 prop_binSR_electron2017_bin3
102 prop_binSR_electron2016M_bin5
103 prop_binSR_electron2016M_bin12
104 prop_binCRF_muon2018_bin3
105 prop_binSR_electron2016M_bin3
106 prop_binCRF_muon2018_bin5
107 prop_binSR_electron2016M_bin8
108 prop_binCRF_muon2018_bin4
109 prop_binSR_electron2016M_bin6
110 prop_binSR_electron2017_bin2
111 QCDScale_VG
112 prop_binCRF_muon2017_bin3
113 prop_binSR_electron2016M_bin4
114 prop_binSR_electron2018_bin12_ZZtoLep
115 prop_binSR_muon2016M_bin10_TTTo2L2Nu
116 prop_binCROS_electron2018_bin4
117 prop_binCRF_muon2018_bin6
118 PF
119 prop_binCRTT_electron2018_bin1
120 prop_binCRF_muon2018_bin7
−2
−1
0
1
2
−0.004
−0.002
0
0.002
0.004
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



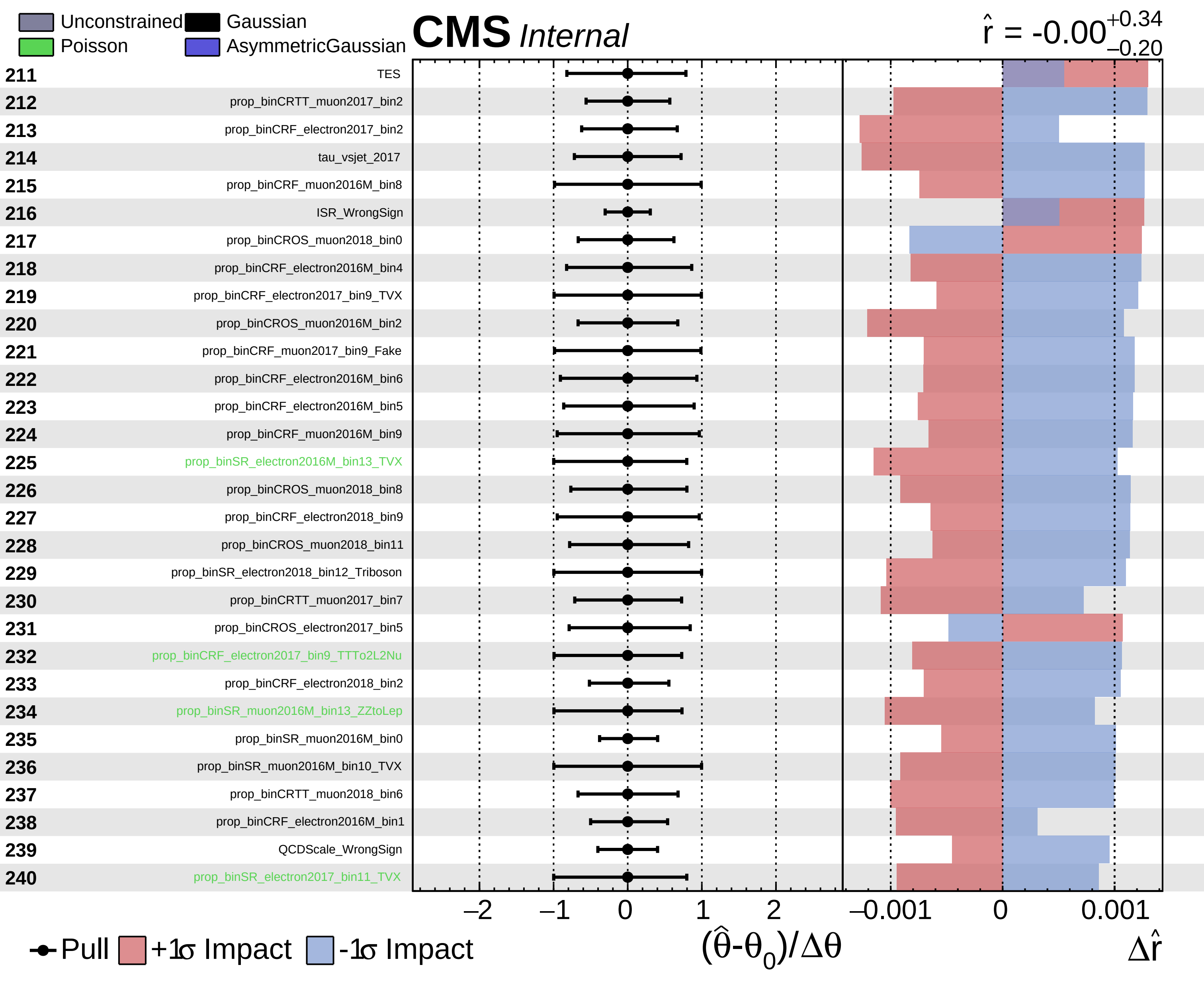
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = -0.00 +0.34 −0.20
121 prop_binCROS_electron2018_bin5
122 prop_binCRF_electron2018_bin4
123 prop_binCROS_muon2017_bin4
124 prop_binCROS_electron2018_bin3
125 prop_binSR_muon2016M_bin13_Fake
126 prop_binCRF_muon2018_bin8
127 prop_binCRF_electron2016M_bin7_WrongSign
128 prop_binCRF_electron2018_bin0
129 prop_binCRF_muon2017_bin2
130 prop_binCRF_muon2017_bin4
131 prop_binCROS_electron2017_bin1
132 prop_binSR_electron2018_bin1
133 prop_binCRF_electron2018_bin6
134 prop_binSR_muon2017_bin11_TVX
135 prop_binCRF_muon2016M_bin3
136 prop_binSR_electron2016M_bin2
137 prop_binCRF_muon2017_bin5
138 prop_binCRF_electron2018_bin3
139 prop_binCRF_muon2016M_bin5
140 pdf_total_Triboson
141 prop_binCRF_muon2018_bin9
142 prop_binCRF_muon2017_bin7
143 prop_binCRF_muon2016M_bin2
144 prop_binCRF_electron2017_bin8_Fake
145 prop_binCRF_electron2017_bin3
146 prop_binCROS_electron2016M_bin2
147 prop_binCROS_muon2018_bin4
148 prop_binCRF_electron2018_bin5
149 prop_binCRF_muon2017_bin1
150 prop_binCROS_electron2018_bin8
−2 −1 0 1 2
−0.002 0 0.002
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



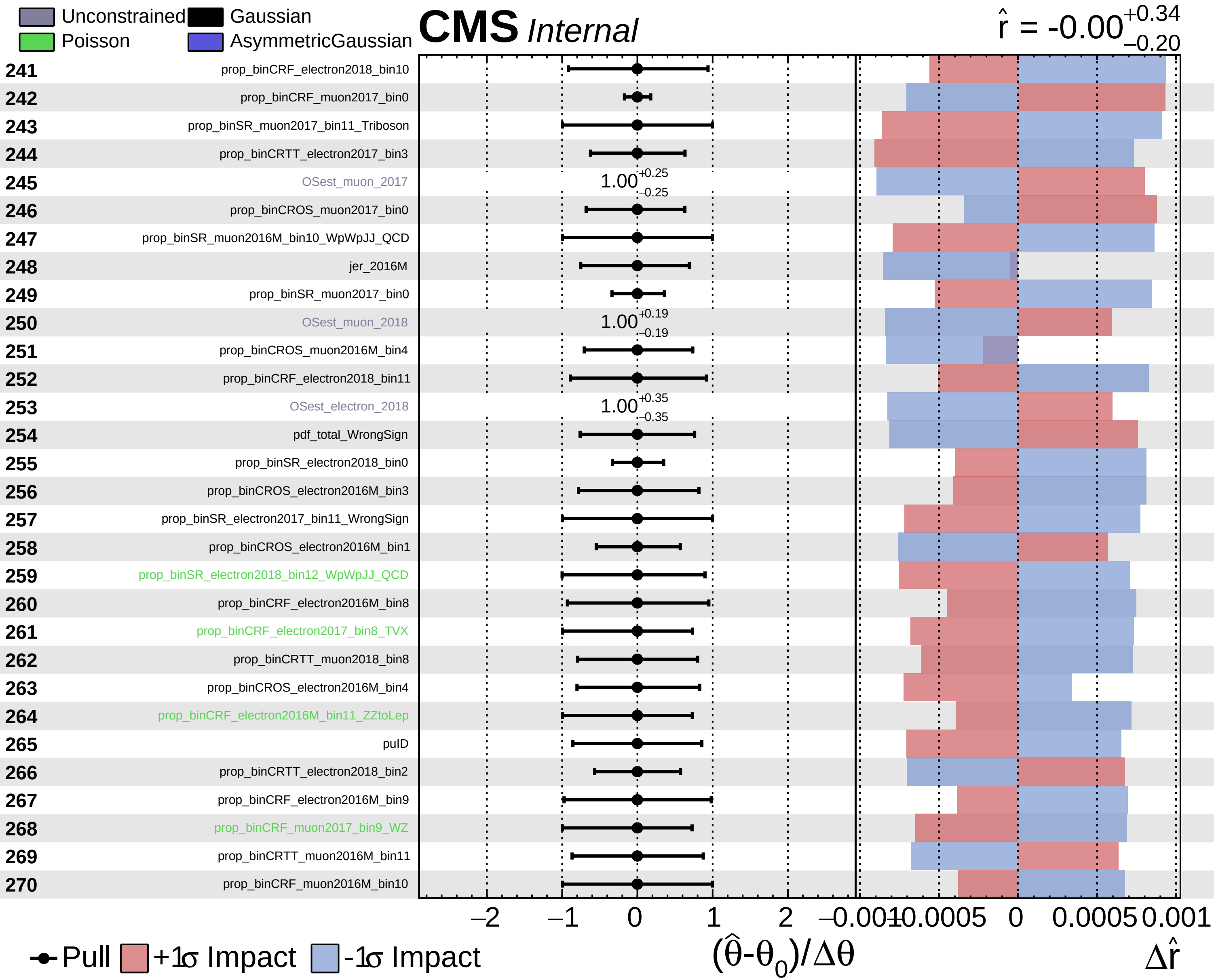
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.34}_{-0.20}$
151 prop_binCRTT_muon2017_bin0
152 prop_binCRF_electron2018_bin7
153 prop_binCROS_muon2018_bin5
154 prop_binCROS_electron2016M_bin5_WrongSign
155 prop_binCRTT_muon2017_bin1
156 prop_binCRF_muon2017_bin6
157 prop_binCRF_muon2016M_bin4
158 prop_binCRTT_muon2017_bin6
159 prop_binCROS_electron2018_bin0
160 prop_binSR_electron2017_bin1
161 prop_binCROS_electron2018_bin7
162 prop_binCRF_muon2016M_bin1
163 prop_binCRF_muon2016M_bin6
164 prop_binCRF_electron2016M_bin2
165 prop_binCROS_muon2018_bin2
166 prop_binCROS_muon2017_bin5
167 prop_binCROS_muon2016M_bin5_WrongSign
168 prop_binCROS_electron2018_bin6
169 prop_binCRF_electron2017_bin4
170 prop_binCRTT_muon2016M_bin0
171 prop_binCRTT_electron2018_bin0
172 prop_binCRF_electron2017_bin7
173 OSest_muon_2016M
$1.00^{+0.30}_{-0.30}$
174 prop_binCROS_electron2017_bin0
175 prop_binCROS_electron2018_bin1
176 tau_vsjet_2016M
177 prop_binCRF_electron2017_bin0
178 prop_binCRF_muon2018_bin0
179 prop_binCROS_electron2018_bin11_WrongSign
180 prop_binCRF_electron2018_bin8
−2 −1 0 1 2
−0.002 −0.001 0 0.001 0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



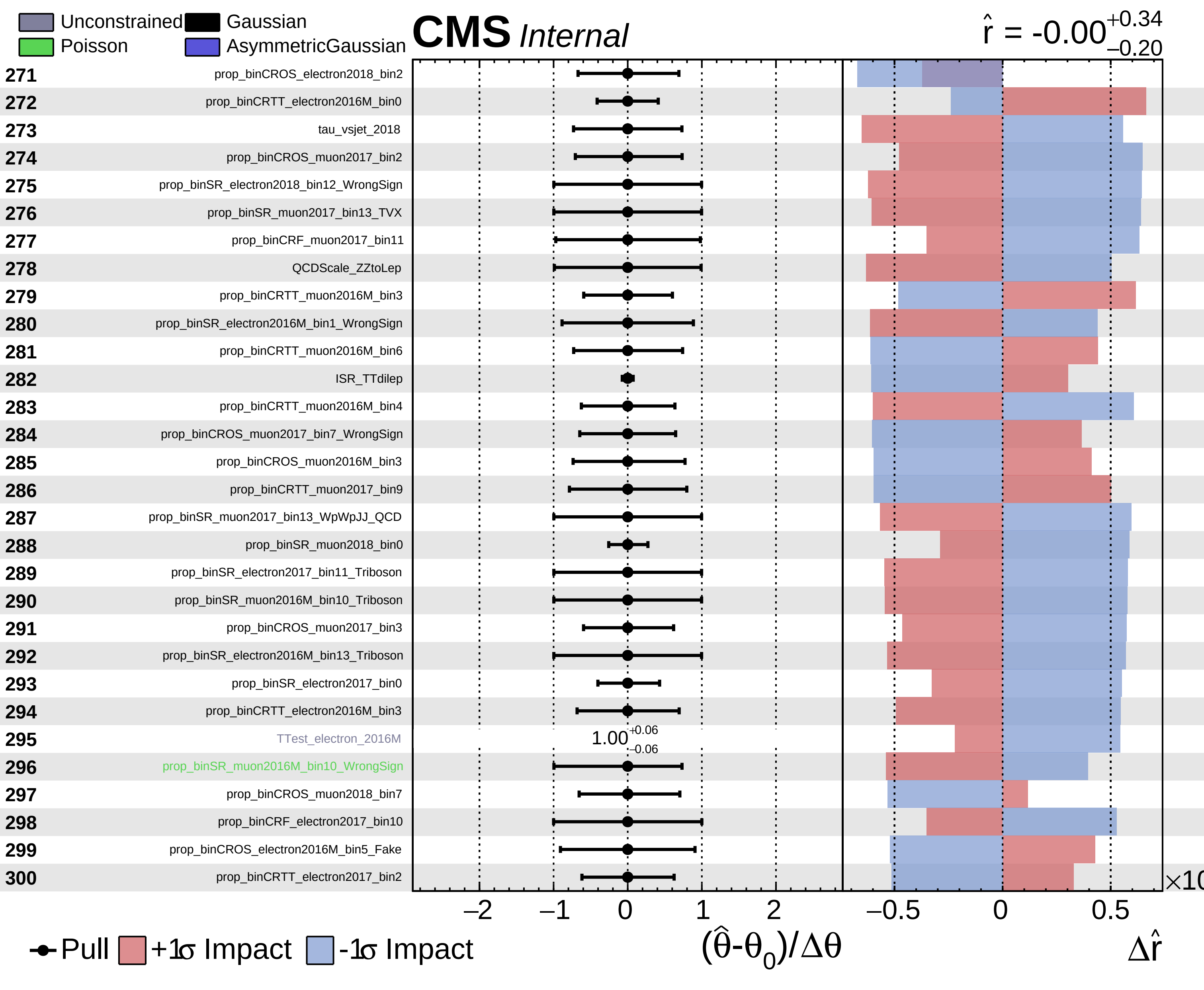
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.34}_{-0.20}$
181 prop_binSR_muon2017_bin13_ZZtoLep
182 prop_binCRTT_electron2016M_bin1
183 prop_binCRF_muon2018_bin1
184 pdf_total_TVX
185 prop_binCRF_muon2017_bin8
186 prop_binCROS_electron2016M_bin6
187 prop_binCRF_electron2017_bin5
188 prop_binSR_muon2017_bin11_WpWpJJ_QCD
189 prop_binCRTT_electron2018_bin5
190 prop_binCROS_electron2017_bin6_Fake
191 prop_binCRF_muon2018_bin10
192 prop_binCROS_electron2017_bin7_WrongSign
193 prop_binCRF_electron2017_bin6
194 prop_binCROS_electron2018_bin9
195 prop_binCROS_electron2016M_bin7_WrongSign
196 prop_binSR_electron2017_bin11_WpWpJJ_QCD
197 prop_binSR_electron2018_bin12_TVX
198 prop_binCROS_muon2018_bin6
199 prop_binCROS_muon2017_bin6
200 prop_binSR_electron2016M_bin1_Fake
201 prop_binCRF_muon2016M_bin7
202 prop_binCRF_electron2016M_bin0
203 prop_binCROS_electron2016M_bin0
204 prop_binCROS_muon2017_bin8
205 prop_binCRTT_muon2018_bin7
206 mistag
207 prop_binCRF_electron2016M_bin3
208 prop_binCRF_electron2017_bin1
209 prop_binCROS_electron2017_bin6_WrongSign
210 FSR_WrongSign
−2
−1
0
1
2
−0.001
0
0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

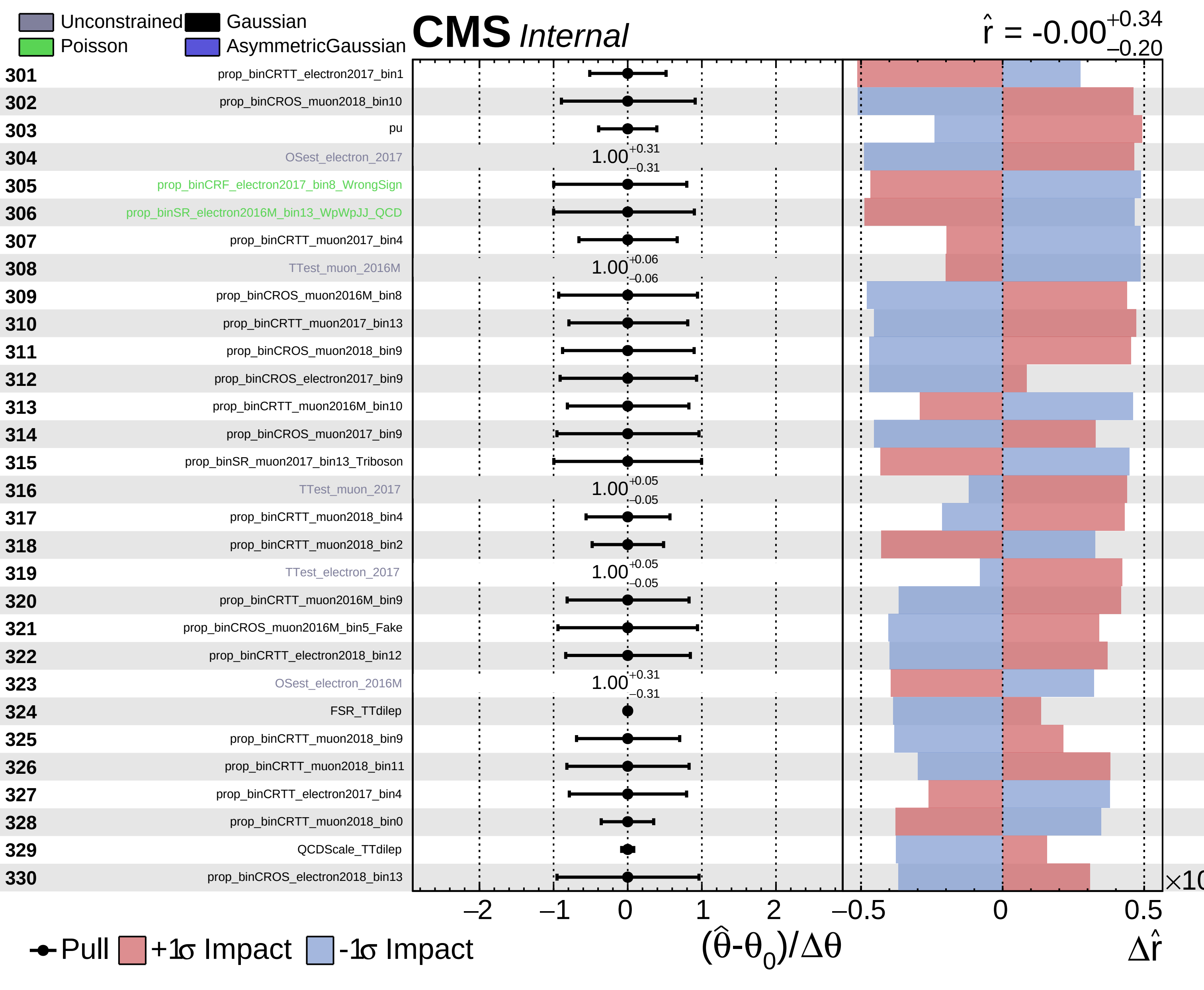


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.34}_{-0.20}$
211 TES
212 prop_binCRTT_muon2017_bin2
213 prop_binCRF_electron2017_bin2
214 tau_vsjet_2017
215 prop_binCRF_muon2016M_bin8
216 ISR_WrongSign
217 prop_binCROS_muon2018_bin0
218 prop_binCRF_electron2016M_bin4
219 prop_binCRF_electron2017_bin9_TVX
220 prop_binCROS_muon2016M_bin2
221 prop_binCRF_muon2017_bin9_Fake
222 prop_binCRF_electron2016M_bin6
223 prop_binCRF_electron2016M_bin5
224 prop_binCRF_muon2016M_bin9
225 prop_binSR_electron2016M_bin13_TVX
226 prop_binCROS_muon2018_bin8
227 prop_binCRF_electron2018_bin9
228 prop_binCROS_muon2018_bin11
229 prop_binSR_electron2018_bin12_Triboson
230 prop_binCRTT_muon2017_bin7
231 prop_binCROS_electron2017_bin5
232 prop_binCRF_electron2017_bin9_TTTo2L2Nu
233 prop_binCRF_electron2018_bin2
234 prop_binSR_muon2016M_bin13_ZZtoLep
235 prop_binSR_muon2016M_bin0
236 prop_binSR_muon2016M_bin10_TVX
237 prop_binCRTT_muon2018_bin6
238 prop_binCRF_electron2016M_bin1
239 QCDScale_WrongSign
240 prop_binSR_electron2017_bin11_TVX
−2 −1 0 1 2
−0.001 0 0.001
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

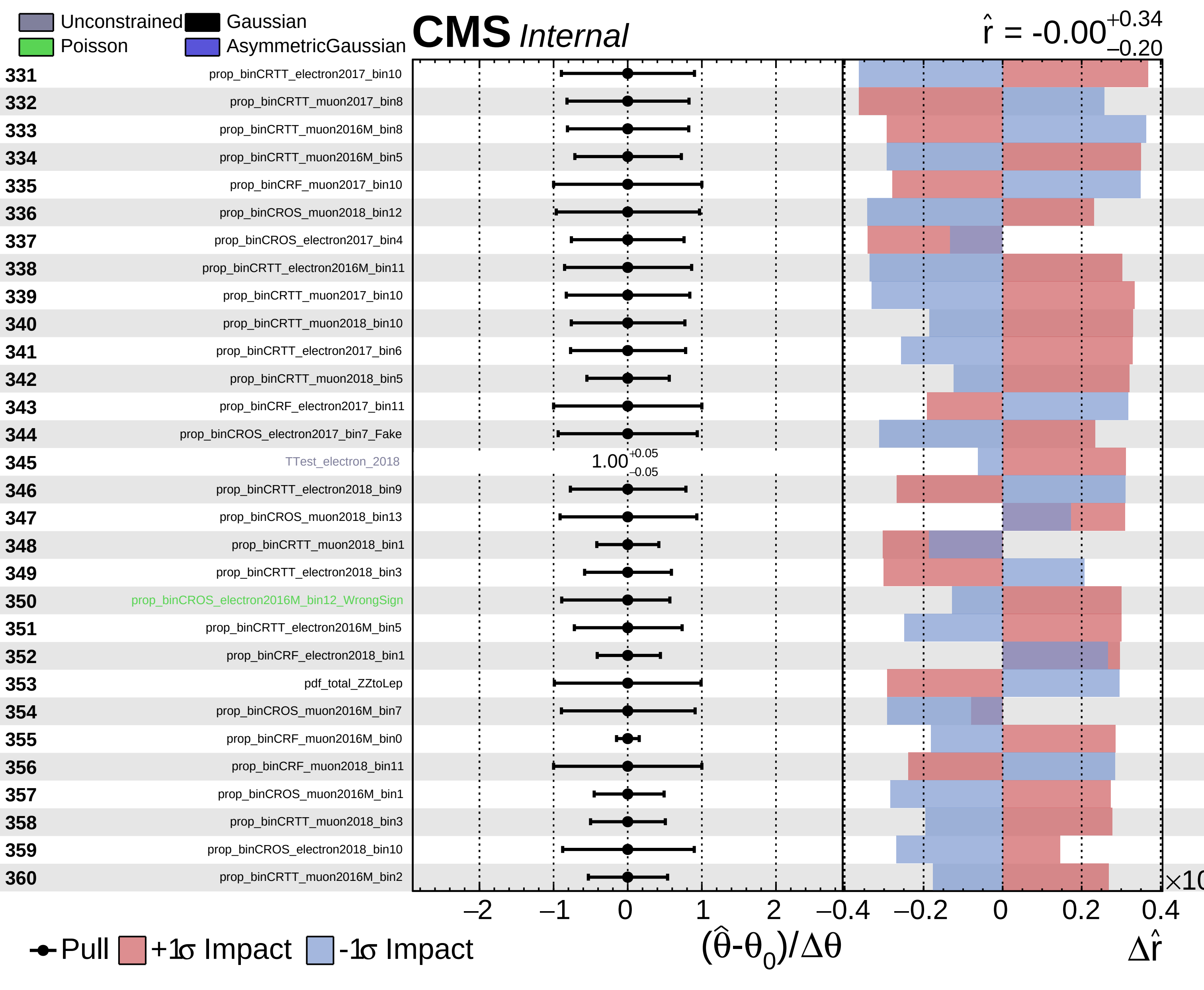


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.34}_{-0.20}$
241 prop_binCRF_electron2018_bin10
242 prop_binCRF_muon2017_bin0
243 prop_binSR_muon2017_bin11_Triboson
244 prop_binCRTT_electron2017_bin3
245 OSest_muon_2017 $1.00^{+0.25}_{-0.25}$
246 prop_binCROS_muon2017_bin0
247 prop_binSR_muon2016M_bin10_WpWpJJ_QCD
248 jer_2016M
249 prop_binSR_muon2017_bin0
250 OSest_muon_2018 $1.00^{+0.19}_{-0.19}$
251 prop_binCROS_muon2016M_bin4
252 prop_binCRF_electron2018_bin11
253 OSest_electron_2018 $1.00^{+0.35}_{-0.35}$
254 pdf_total_WrongSign
255 prop_binSR_electron2018_bin0
256 prop_binCROS_electron2016M_bin3
257 prop_binSR_electron2017_bin11_WrongSign
258 prop_binCROS_electron2016M_bin1
259 prop_binSR_electron2018_bin12_WpWpJJ_QCD
260 prop_binCRF_electron2016M_bin8
261 prop_binCRF_electron2017_bin8_TVX
262 prop_binCRTT_muon2018_bin8
263 prop_binCROS_electron2016M_bin4
264 prop_binCRF_electron2016M_bin11_ZZtoLep
265 puID
266 prop_binCRTT_electron2018_bin2
267 prop_binCRF_electron2016M_bin9
268 prop_binCRF_muon2017_bin9_WZ
269 prop_binCRTT_muon2016M_bin11
270 prop_binCRF_muon2016M_bin10
−2 −1 0 1 2
−0.001 −0.0005 0 0.0005 0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

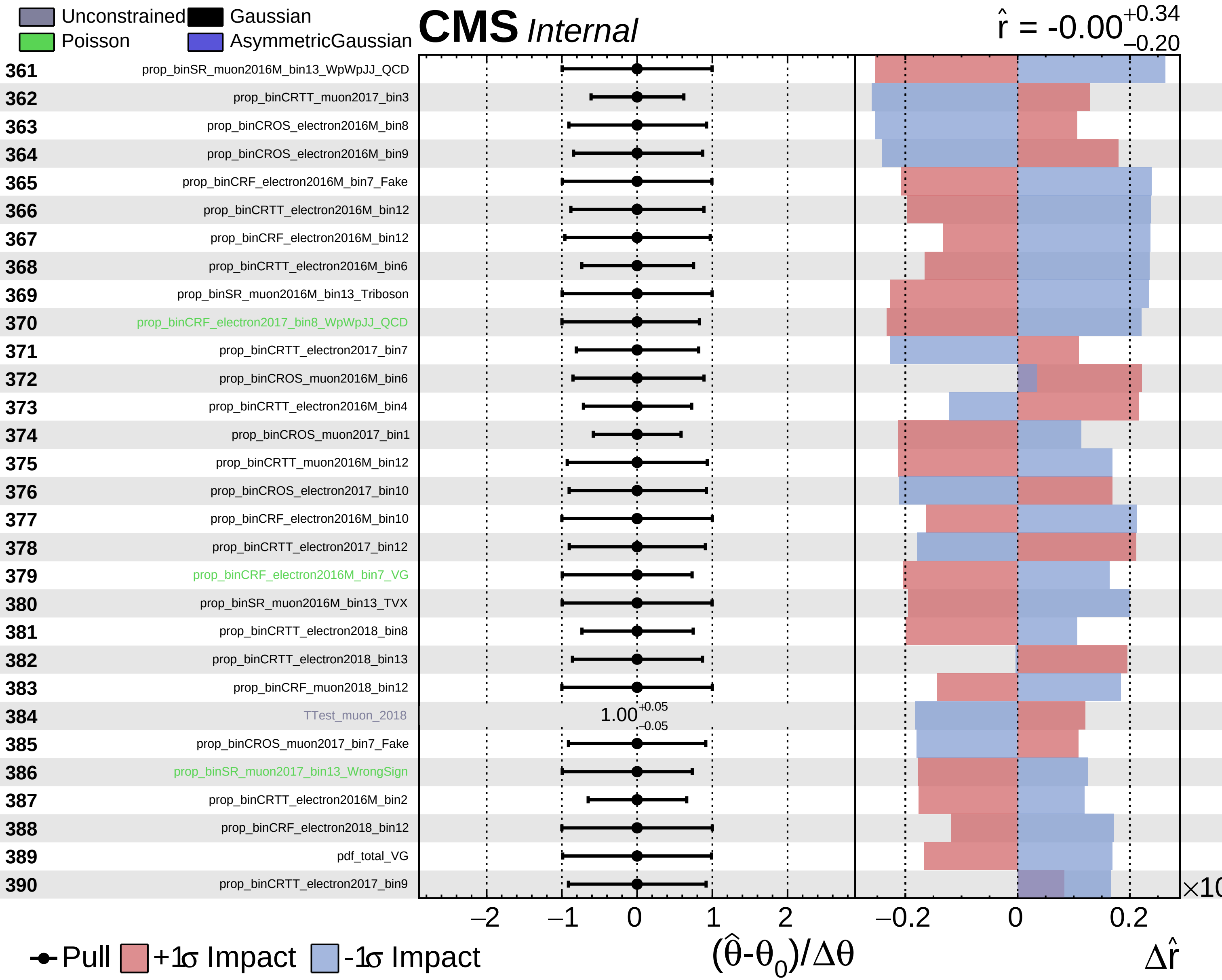

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = -0.00 +0.34 −0.20
271 prop_binCROS_electron2018_bin2
272 prop_binCRTT_electron2016M_bin0
273 tau_vsjet_2018
274 prop_binCROS_muon2017_bin2
275 prop_binSR_electron2018_bin12_WrongSign
276 prop_binSR_muon2017_bin13_TVX
277 prop_binCRF_muon2017_bin11
278 QCDScale_ZZtoLep
279 prop_binCRTT_muon2016M_bin3
280 prop_binSR_electron2016M_bin1_WrongSign
281 prop_binCRTT_muon2016M_bin6
282 ISR_TTdilep
283 prop_binCRTT_muon2016M_bin4
284 prop_binCROS_muon2017_bin7_WrongSign
285 prop_binCROS_muon2016M_bin3
286 prop_binCRTT_muon2017_bin9
287 prop_binSR_muon2017_bin13_WpWpJJ_QCD
288 prop_binSR_muon2018_bin0
289 prop_binSR_electron2017_bin11_Triboson
290 prop_binSR_muon2016M_bin10_Triboson
291 prop_binCROS_muon2017_bin3
292 prop_binSR_electron2016M_bin13_Triboson
293 prop_binSR_electron2017_bin0
294 prop_binCRTT_electron2016M_bin3
295 TTest_electron_2016M
1.00 +0.06 −0.06
296 prop_binSR_muon2016M_bin10_WrongSign
297 prop_binCROS_muon2018_bin7
298 prop_binCRF_electron2017_bin10
299 prop_binCROS_electron2016M_bin5_Fake
300 prop_binCRTT_electron2017_bin2
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂



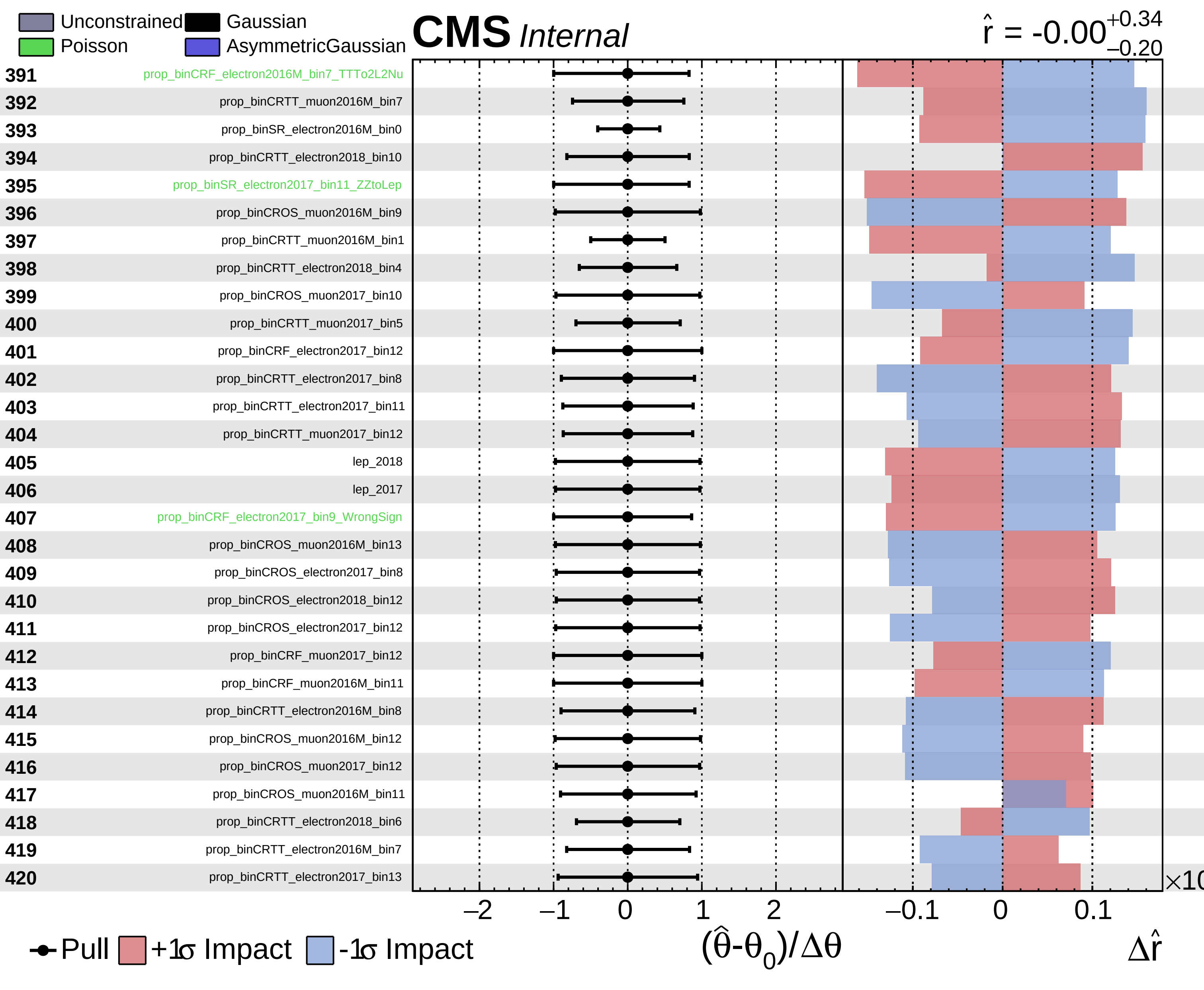
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = -0.00 +0.34 −0.20
301 prop_binCRTT_electron2017_bin1
302 prop_binCROS_muon2018_bin10
303 pu
304 OSest_electron_2017 1.00 +0.31 −0.31
305 prop_binCRF_electron2017_bin8_WrongSign
306 prop_binSR_electron2016M_bin13_WpWpJJ_QCD
307 prop_binCRTT_muon2017_bin4
308 TTest_muon_2016M 1.00 +0.06 −0.06
309 prop_binCROS_muon2016M_bin8
310 prop_binCRTT_muon2017_bin13
311 prop_binCROS_muon2018_bin9
312 prop_binCROS_electron2017_bin9
313 prop_binCRTT_muon2016M_bin10
314 prop_binCROS_muon2017_bin9
315 prop_binSR_muon2017_bin13_Triboson
316 TTest_muon_2017 1.00 +0.05 −0.05
317 prop_binCRTT_muon2018_bin4
318 prop_binCRTT_muon2018_bin2
319 TTest_electron_2017 1.00 +0.05 −0.05
320 prop_binCRTT_muon2016M_bin9
321 prop_binCROS_muon2016M_bin5_Fake
322 prop_binCRTT_electron2018_bin12
323 OSest_electron_2016M 1.00 +0.31 −0.31
324 FSR_TTdilep
325 prop_binCRTT_muon2018_bin9
326 prop_binCRTT_muon2018_bin11
327 prop_binCRTT_electron2017_bin4
328 prop_binCRTT_muon2018_bin0
329 QCDScale_TTdilep
330 prop_binCROS_electron2018_bin13
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂

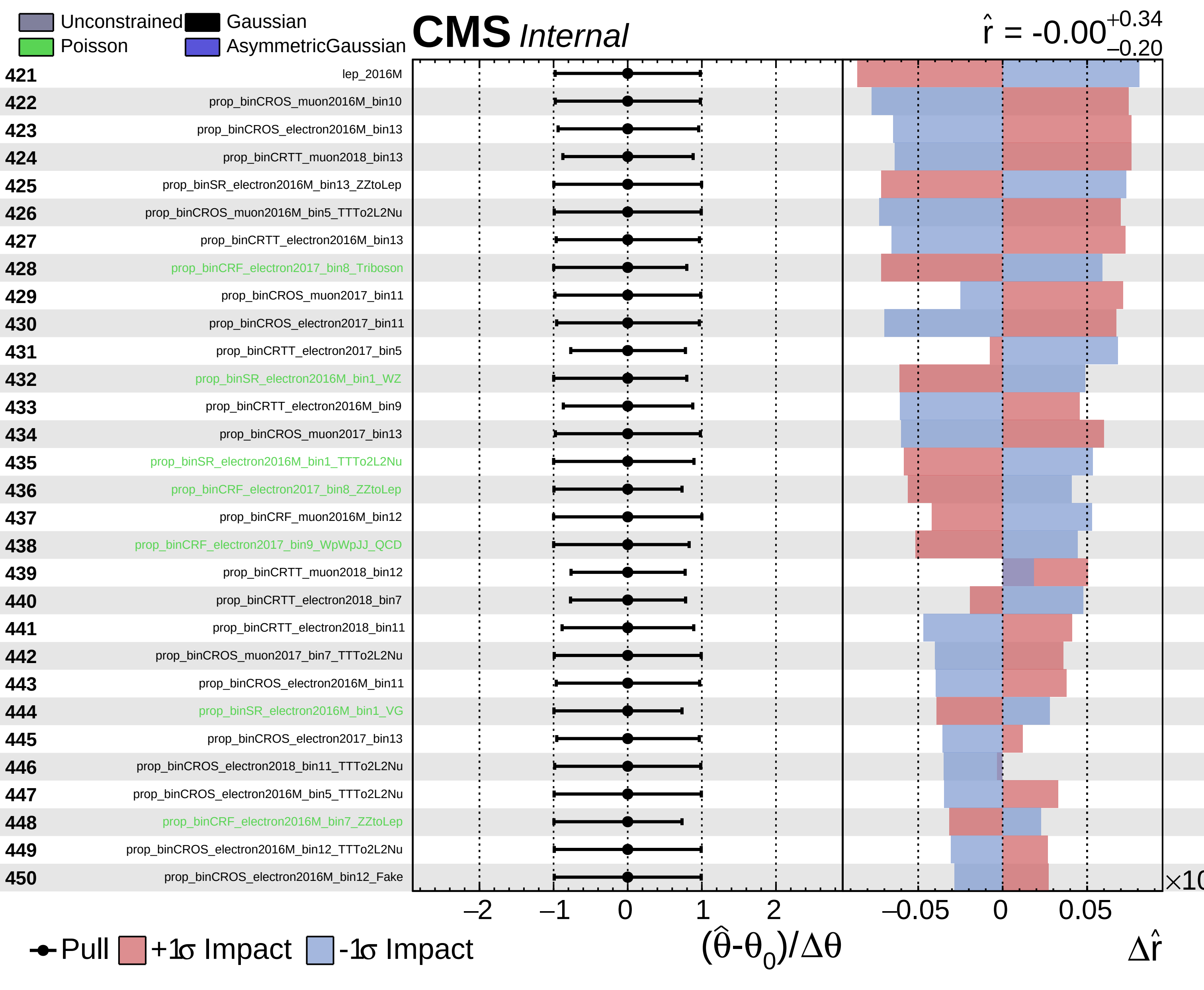


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = -0.00 +0.34 −0.20
331 prop_binCRTT_electron2017_bin10
332 prop_binCRTT_muon2017_bin8
333 prop_binCRTT_muon2016M_bin8
334 prop_binCRTT_muon2016M_bin5
335 prop_binCRF_muon2017_bin10
336 prop_binCROS_muon2018_bin12
337 prop_binCROS_electron2017_bin4
338 prop_binCRTT_electron2016M_bin11
339 prop_binCRTT_muon2017_bin10
340 prop_binCRTT_muon2018_bin10
341 prop_binCRTT_electron2017_bin6
342 prop_binCRTT_muon2018_bin5
343 prop_binCRF_electron2017_bin11
344 prop_binCROS_electron2017_bin7_Fake
345 TTest_electron_2018
1.00 +0.05 −0.05
346 prop_binCRTT_electron2018_bin9
347 prop_binCROS_muon2018_bin13
348 prop_binCRTT_muon2018_bin1
349 prop_binCRTT_electron2018_bin3
350 prop_binCROS_electron2016M_bin12_WrongSign
351 prop_binCRTT_electron2016M_bin5
352 prop_binCRF_electron2018_bin1
353 pdf_total_ZZtoLep
354 prop_binCROS_muon2016M_bin7
355 prop_binCRF_muon2016M_bin0
356 prop_binCRF_muon2018_bin11
357 prop_binCROS_muon2016M_bin1
358 prop_binCRTT_muon2018_bin3
359 prop_binCROS_electron2018_bin10
360 prop_binCRTT_muon2016M_bin2
−2 −1 0 1 2
−0.4 −0.2 0 0.2 0.4
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

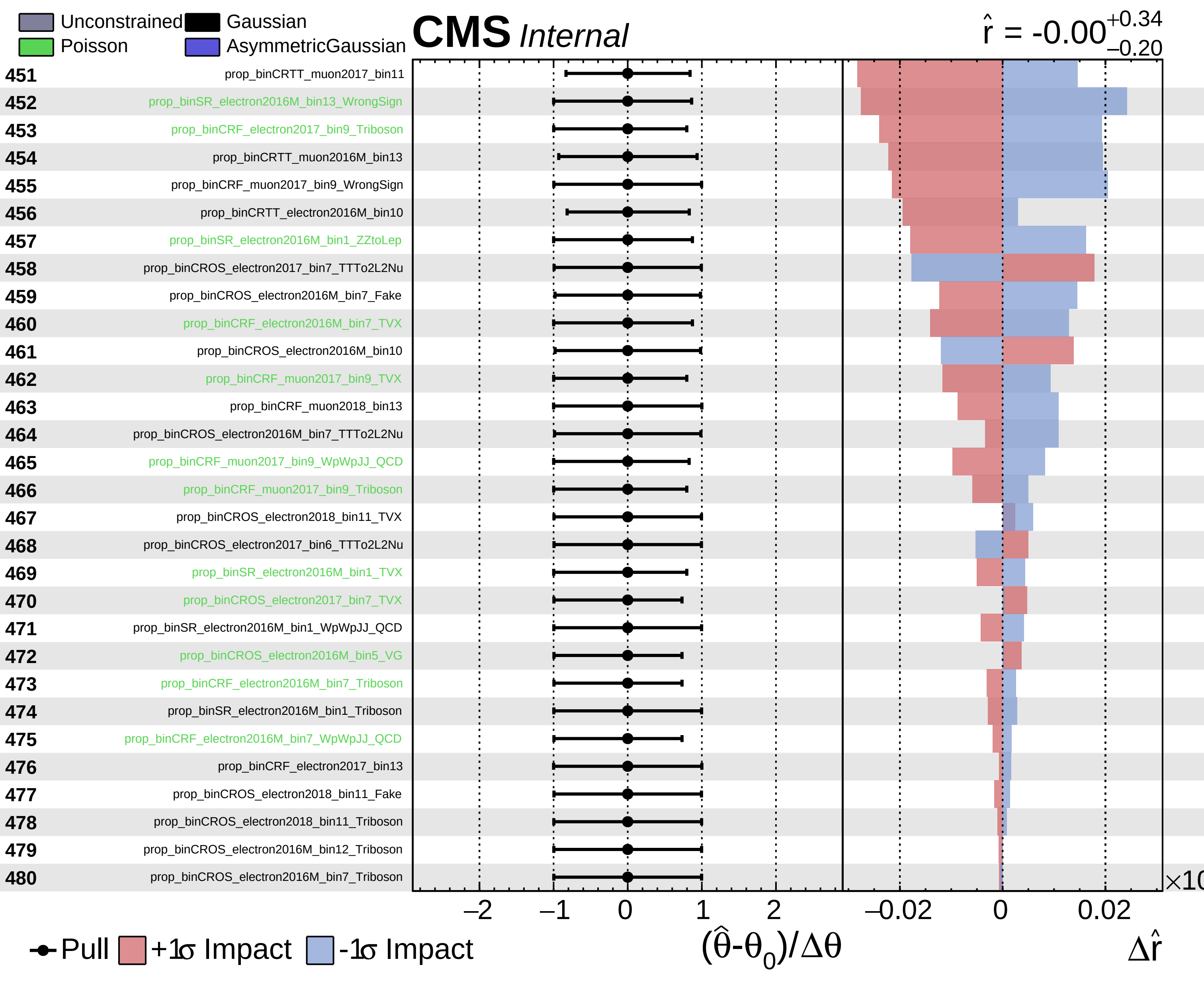


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.34}_{-0.20}$
361 prop_binSR_muon2016M_bin13_WpWpJJ_QCD
362 prop_binCRTT_muon2017_bin3
363 prop_binCROS_electron2016M_bin8
364 prop_binCROS_electron2016M_bin9
365 prop_binCRF_electron2016M_bin7_Fake
366 prop_binCRTT_electron2016M_bin12
367 prop_binCRF_electron2016M_bin12
368 prop_binCRTT_electron2016M_bin6
369 prop_binSR_muon2016M_bin13_Triboson
370 prop_binCRF_electron2017_bin8_WpWpJJ_QCD
371 prop_binCRTT_electron2017_bin7
372 prop_binCROS_muon2016M_bin6
373 prop_binCRTT_electron2016M_bin4
374 prop_binCROS_muon2017_bin1
375 prop_binCRTT_muon2016M_bin12
376 prop_binCROS_electron2017_bin10
377 prop_binCRF_electron2016M_bin10
378 prop_binCRTT_electron2017_bin12
379 prop_binCRF_electron2016M_bin7_VG
380 prop_binSR_muon2016M_bin13_TVX
381 prop_binCRTT_electron2018_bin8
382 prop_binCRTT_electron2018_bin13
383 prop_binCRF_muon2018_bin12
384 TTest_muon_2018
$1.00^{+0.05}_{-0.05}$
385 prop_binCROS_muon2017_bin7_Fake
386 prop_binSR_muon2017_bin13_WrongSign
387 prop_binCRTT_electron2016M_bin2
388 prop_binCRF_electron2018_bin12
389 pdf_total_VG
390 prop_binCRTT_electron2017_bin9
−2 −1 0 1 2
−0.2 0 0.2
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

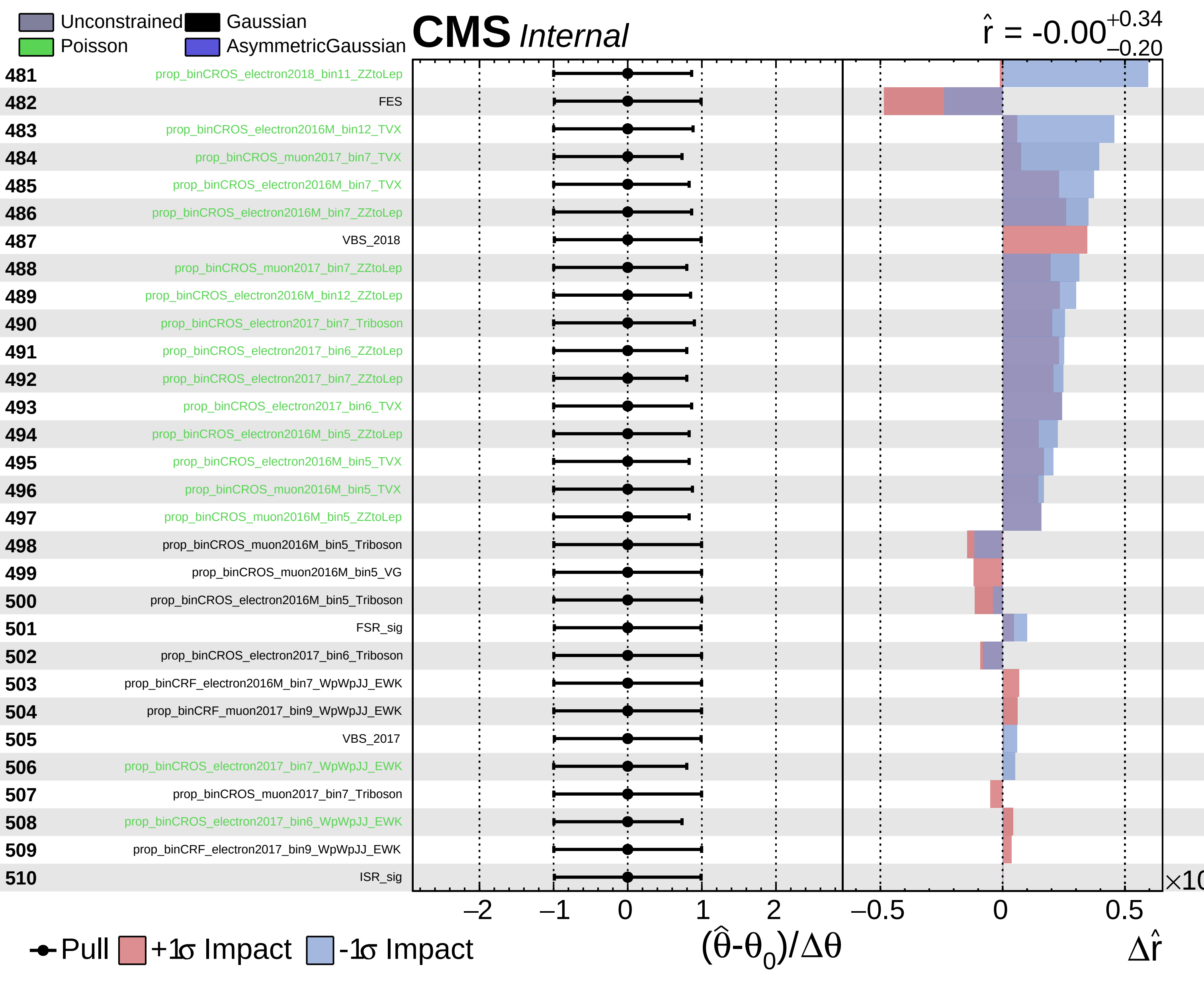

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.34}_{-0.20}$
391 prop_binCRF_electron2016M_bin7_TTTo2L2Nu
392 prop_binCRTT_muon2016M_bin7
393 prop_binSR_electron2016M_bin0
394 prop_binCRTT_electron2018_bin10
395 prop_binSR_electron2017_bin11_ZZtoLep
396 prop_binCROS_muon2016M_bin9
397 prop_binCRTT_muon2016M_bin1
398 prop_binCRTT_electron2018_bin4
399 prop_binCROS_muon2017_bin10
400 prop_binCRTT_muon2017_bin5
401 prop_binCRF_electron2017_bin12
402 prop_binCRTT_electron2017_bin8
403 prop_binCRTT_electron2017_bin11
404 prop_binCRTT_muon2017_bin12
405 lep_2018
406 lep_2017
407 prop_binCRF_electron2017_bin9_WrongSign
408 prop_binCROS_muon2016M_bin13
409 prop_binCROS_electron2017_bin8
410 prop_binCROS_electron2018_bin12
411 prop_binCROS_electron2017_bin12
412 prop_binCRF_muon2017_bin12
413 prop_binCRF_muon2016M_bin11
414 prop_binCRTT_electron2016M_bin8
415 prop_binCROS_muon2016M_bin12
416 prop_binCROS_muon2017_bin12
417 prop_binCROS_muon2016M_bin11
418 prop_binCRTT_electron2018_bin6
419 prop_binCRTT_electron2016M_bin7
420 prop_binCRTT_electron2017_bin13
−2
−1
0
1
2
−0.1
0
0.1
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



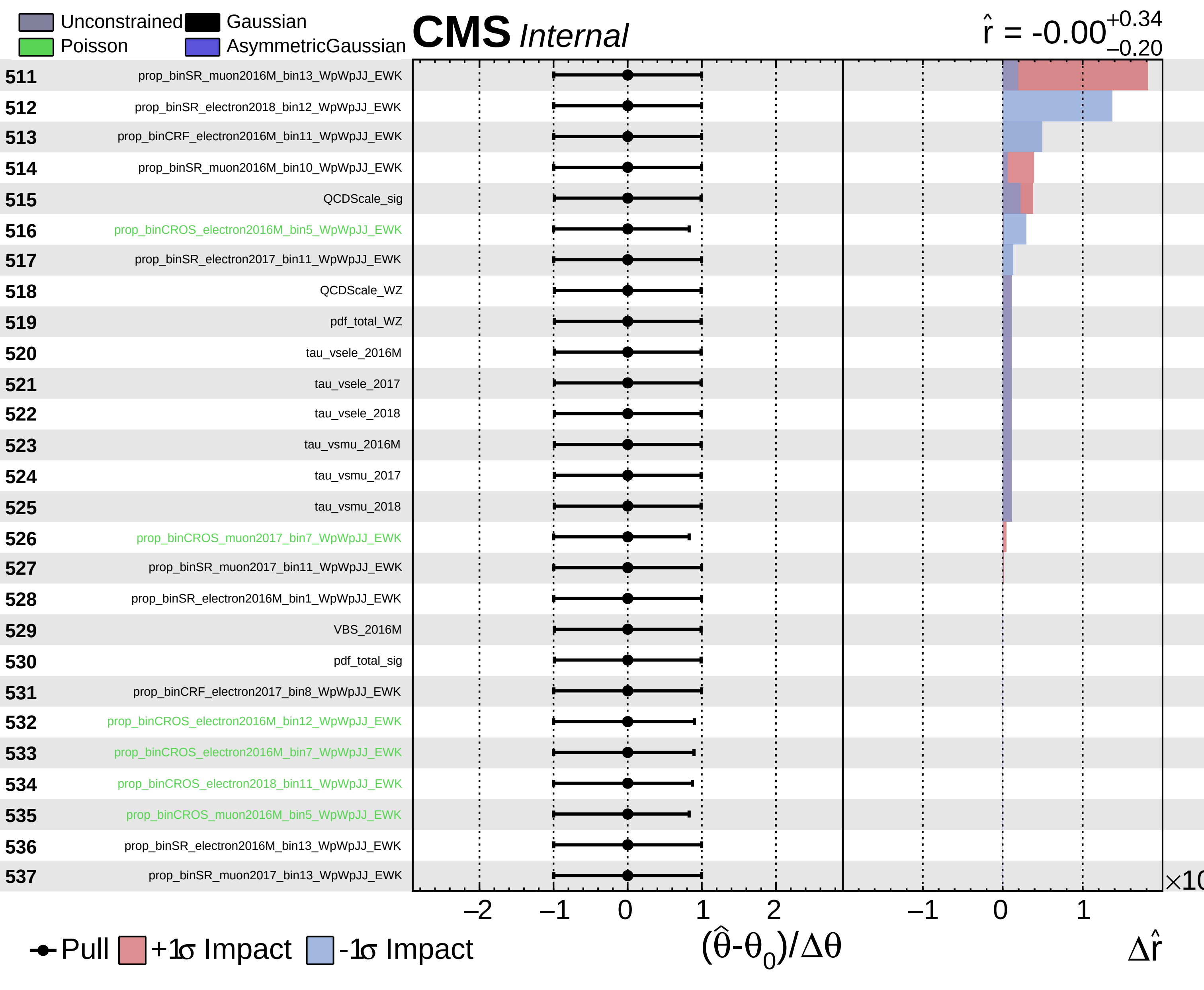
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.34}_{-0.20}$
421 lep_2016M
422 prop_binCROS_muon2016M_bin10
423 prop_binCROS_electron2016M_bin13
424 prop_binCRTT_muon2018_bin13
425 prop_binSR_electron2016M_bin13_ZZtoLep
426 prop_binCROS_muon2016M_bin5_TTTo2L2Nu
427 prop_binCRTT_electron2016M_bin13
428 prop_binCRF_electron2017_bin8_Triboson
429 prop_binCROS_muon2017_bin11
430 prop_binCROS_electron2017_bin11
431 prop_binCRTT_electron2017_bin5
432 prop_binSR_electron2016M_bin1_WZ
433 prop_binCRTT_electron2016M_bin9
434 prop_binCROS_muon2017_bin13
435 prop_binSR_electron2016M_bin1_TTTo2L2Nu
436 prop_binCRF_electron2017_bin8_ZZtoLep
437 prop_binCRF_muon2016M_bin12
438 prop_binCRF_electron2017_bin9_WpWpJJ_QCD
439 prop_binCRTT_muon2018_bin12
440 prop_binCRTT_electron2018_bin7
441 prop_binCRTT_electron2018_bin11
442 prop_binCROS_muon2017_bin7_TTTo2L2Nu
443 prop_binCROS_electron2016M_bin11
444 prop_binSR_electron2016M_bin1_VG
445 prop_binCROS_electron2017_bin13
446 prop_binCROS_electron2018_bin11_TTTo2L2Nu
447 prop_binCROS_electron2016M_bin5_TTTo2L2Nu
448 prop_binCRF_electron2016M_bin7_ZZtoLep
449 prop_binCROS_electron2016M_bin12_TTTo2L2Nu
450 prop_binCROS_electron2016M_bin12_Fake
−2 −1 0 1 2
−0.05 0 0.05
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.34}_{-0.20}$
451 prop_binCRTT_muon2017_bin11
452 prop_binSR_electron2016M_bin13_WrongSign
453 prop_binCRF_electron2017_bin9_Triboson
454 prop_binCRTT_muon2016M_bin13
455 prop_binCRF_muon2017_bin9_WrongSign
456 prop_binCRTT_electron2016M_bin10
457 prop_binSR_electron2016M_bin1_ZZtoLep
458 prop_binCROS_electron2017_bin7_TTTo2L2Nu
459 prop_binCROS_electron2016M_bin7_Fake
460 prop_binCRF_electron2016M_bin7_TVX
461 prop_binCROS_electron2016M_bin10
462 prop_binCRF_muon2017_bin9_TVX
463 prop_binCRF_muon2018_bin13
464 prop_binCROS_electron2016M_bin7_TTTo2L2Nu
465 prop_binCRF_muon2017_bin9_WpWpJJ_QCD
466 prop_binCRF_muon2017_bin9_Triboson
467 prop_binCROS_electron2018_bin11_TVX
468 prop_binCROS_electron2017_bin6_TTTo2L2Nu
469 prop_binSR_electron2016M_bin1_TVX
470 prop_binCROS_electron2017_bin7_TVX
471 prop_binSR_electron2016M_bin1_WpWpJJ_QCD
472 prop_binCROS_electron2016M_bin5_VG
473 prop_binCRF_electron2016M_bin7_Triboson
474 prop_binSR_electron2016M_bin1_Triboson
475 prop_binCRF_electron2016M_bin7_WpWpJJ_QCD
476 prop_binCRF_electron2017_bin13
477 prop_binCROS_electron2018_bin11_Fake
478 prop_binCROS_electron2018_bin11_Triboson
479 prop_binCROS_electron2016M_bin12_Triboson
480 prop_binCROS_electron2016M_bin7_Triboson
−2
−1
0
1
2
−0.02
0
0.02
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.34}_{-0.20}$
481 prop_binCROS_electron2018_bin11_ZZtoLep
482 FES
483 prop_binCROS_electron2016M_bin12_TVX
484 prop_binCROS_muon2017_bin7_TVX
485 prop_binCROS_electron2016M_bin7_TVX
486 prop_binCROS_electron2016M_bin7_ZZtoLep
487 VBS_2018
488 prop_binCROS_muon2017_bin7_ZZtoLep
489 prop_binCROS_electron2016M_bin12_ZZtoLep
490 prop_binCROS_electron2017_bin7_Triboson
491 prop_binCROS_electron2017_bin6_ZZtoLep
492 prop_binCROS_electron2017_bin7_ZZtoLep
493 prop_binCROS_electron2017_bin6_TVX
494 prop_binCROS_electron2016M_bin5_ZZtoLep
495 prop_binCROS_electron2016M_bin5_TVX
496 prop_binCROS_muon2016M_bin5_TVX
497 prop_binCROS_muon2016M_bin5_ZZtoLep
498 prop_binCROS_muon2016M_bin5_Triboson
499 prop_binCROS_muon2016M_bin5_VG
500 prop_binCROS_electron2016M_bin5_Triboson
501 FSR_sig
502 prop_binCROS_electron2017_bin6_Triboson
503 prop_binCRF_electron2016M_bin7_WpWpJJ_EWK
504 prop_binCRF_muon2017_bin9_WpWpJJ_EWK
505 VBS_2017
506 prop_binCROS_electron2017_bin7_WpWpJJ_EWK
507 prop_binCROS_muon2017_bin7_Triboson
508 prop_binCROS_electron2017_bin6_WpWpJJ_EWK
509 prop_binCRF_electron2017_bin9_WpWpJJ_EWK
510 ISR_sig
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.34}_{-0.20}$
511 prop_binSR_muon2016M_bin13_WpWpJJ_EWK
512 prop_binSR_electron2018_bin12_WpWpJJ_EWK
513 prop_binCRF_electron2016M_bin11_WpWpJJ_EWK
514 prop_binSR_muon2016M_bin10_WpWpJJ_EWK
515 QCDScale_sig
516 prop_binCROS_electron2016M_bin5_WpWpJJ_EWK
517 prop_binSR_electron2017_bin11_WpWpJJ_EWK
518 QCDScale_WZ
519 pdf_total_WZ
520 tau_vsele_2016M
521 tau_vsele_2017
522 tau_vsele_2018
523 tau_vsmu_2016M
524 tau_vsmu_2017
525 tau_vsmu_2018
526 prop_binCROS_muon2017_bin7_WpWpJJ_EWK
527 prop_binSR_muon2017_bin11_WpWpJJ_EWK
528 prop_binSR_electron2016M_bin1_WpWpJJ_EWK
529 VBS_2016M
530 pdf_total_sig
531 prop_binCRF_electron2017_bin8_WpWpJJ_EWK
532 prop_binCROS_electron2016M_bin12_WpWpJJ_EWK
533 prop_binCROS_electron2016M_bin7_WpWpJJ_EWK
534 prop_binCROS_electron2018_bin11_WpWpJJ_EWK
535 prop_binCROS_muon2016M_bin5_WpWpJJ_EWK
536 prop_binSR_electron2016M_bin13_WpWpJJ_EWK
537 prop_binSR_muon2017_bin13_WpWpJJ_EWK
−2 −1 0 1 2
−1 0 1
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$